

Problem Sets Between Test 1 and Test 2

*Only turn in problems that are **not** bracketed.* Bracketed problems are additional problems you can look at. Round brackets indicate problems that may help you with problems that are assigned; square brackets are additional problems on material that you should know, but you are not required to write up solutions; curly brackets are truly optional and may contain extra nuggets that you will not be required to know but may be interested in.

Additional assignments will be filled in over time.

notation	meaning
unbracketed	assigned problem – turn these in for grading
()	helper/warm-up problem
[]	additional problems (you are responsible for content, but don't turn them in)
{ }	covers optional material

Due	Task	Source	Problems
Wed 2/1	PS 1	Rosen 9.1	(1) (3,5,7,9) types of graph 4 types of graph 8 types of graph 11 graphs and relations 13 intersection graph 18 Fred 31 precedence graph
Fri 2/2	WW01	Rosen 9.2	
Mon 2/6	WW02	Rosen 9.3	
Wed 2/8	PS 2	Rosen 9.4 Rosen 9.5	2 paths 6 components 12 strong/weak connectivity 18 isomorphic? 20 isomorphic? 54 farmer puzzle [1] paths [3–5] [11] strong/weak connectivity [19, 21] isomorphic? 2–8 even Euler paths/circuits 10 bridges 18 directed Euler 20 directed Euler 26 which have Euler circuits? [1–7 odd] Euler paths/circuits [19, 21] directed Euler Note: Your answers to homework in this class should typically include some sort of explanation. Answers like “Yes”, “No”, and “17”, are rarely sufficient.
Fri 2/9	PS 3	Rosen 9.7	2–4 planar 12 regions [13] regions 14 vertices
Mon 2/12	PS 4	Rosen 9.RQ Rosen 9.SE	3 degree 3–5 isomorphic?
Wed 2/14	PS 5	Rosen 9.7	6–8 planar 23–25 Kuratowski
Fri 2/16	PS 6	Rosen 9.5 Rosen 9.7	22–23 directed Euler 32–40 even Hamilton 16 bipartite, planar
Mon 2/19	WW03	Rosen 5.1	
Wed 2/21	WW04	Rosen 5.1 – 5.2	

Due	Task	Source	Problems
Fri 2/23	PS 6	Rosen 5.2 Rosen 5.3	4 bowl 18 students 29 initials 40 houses 1 permutations 8 race [11] bits 12 bits [15] letters 18 coins [19] coins 23 line up [24] line up 30 committee
Mon 2/26	WW05	Rosen 5.1–5.4	
Wed 2/28		Rosen 5.RQ Rosen 5.SE Rosen 9.RQ Rosen 9.SE	[1–3] counting principles [5] [6–7] pigeonhole principle [8] combinations and permutations [13] counting problem schemas [15] scrambling letters [1–2] selecting items [3] T/F [4] bitstrings [6] phone numbers [7] ice cream [12] zodiac [15] cards [21] donuts [22] permutations [23] combinations [34] round table [35] advising [38] PEPPERCORN [40] license plates [1] types of graph [2] models [3–5] degree [6–7] graph examples [8] bipartite [9] graph representation [10] isomorphism [11] connected [12] checking isomorphism [13] Euler paths [14a] Hamilton paths [17] planar graph [18] Euler's formula [19] Kuratowski's Theorem [3–5] isomorphic? [21] invariants